

Uncertainty calculations, step by step

Supporting workings for the PHYS 1552 Experiment 1 data analysis exercise (Investigation II and Analysis III). Every number below is computed from the spreadsheet data by *analysis.py*.

Step 1: error bar on each velocity point

Each point is $v_y = (y_2 - y_1) / \Delta t$. Two quantities are uncertain:

- Position: $\delta y = \pm 2$ mm on *each* tracked position (how precisely the puck centre can be clicked in Tracker, about 1–2 pixels).
- Time: $\Delta t = 1/30$ s = 0.0333 s, set by the camera frame rate, so $\delta t \approx 0$ (negligible).

The difference of two independent errors adds in quadrature:

$$\delta(\Delta y) = \sqrt{(\delta y)^2 + (\delta y)^2} = \sqrt{2} \cdot \delta y = 1.414 \times 0.002 \text{ m} = 0.00283 \text{ m}$$

$$\delta v_y = \sqrt{2} \cdot \delta y / \Delta t = 0.00283 / 0.0333 = \pm 0.085 \text{ m s}^{-1}$$

Example (recording 1, first point): $v_y = (0.0424 - 0.00385) / 0.0333 = 1.16 \pm 0.085 \text{ m s}^{-1}$. (Recording 1 has one interval of 0.1 s where frames were skipped; there $\delta v_y = 0.00283/0.1 = 0.028 \text{ m s}^{-1}$.)

Step 2: slope and its fit uncertainty (what Excel LINEST does)

For the N points (t, v_y) of one recording, with means $\bar{t}$ and $\bar{v}$:

$$S_{tt} = \sum (t - \bar{t})^2, \quad S_{tv} = \sum (t - \bar{t})(v - \bar{v}), \quad \text{slope } m = S_{tv} / S_{tt}, \quad \text{intercept } c = \bar{v} - m \bar{t}$$

$$\text{residual standard deviation } s = \sqrt{[\sum (v - (mt + c))^2 / (N - 2)]}, \quad \text{slope uncertainty } \delta m = s / \sqrt{S_{tt}}$$

Worked for recording 1: $m = -27.104 / 44.193 = -0.6133 \text{ m s}^{-2}$; $s = \sqrt{(1.2590 / 76)} = 0.1287 \text{ m s}^{-1}$; $\delta m = 0.1287 / \sqrt{44.193} = 0.1287 / 6.648 = 0.0194 \text{ m s}^{-2}$.

Rec.	N	$S_{tt} (\text{s}^2)$	$S_{tv} (\text{m})$	$\Sigma \text{ residual}^2$	$s (\text{m s}^{-1})$	slope $m (\text{m s}^{-2})$	$\delta m (\text{m s}^{-2})$	$\delta m / m $
1	78	44.193	-27.104	1.2590	0.1287	-0.6133	0.0194	3.16 %
2	65	25.422	-15.796	0.5427	0.0928	-0.6213	0.0184	2.96 %
3	76	40.639	-24.637	0.6264	0.0920	-0.6062	0.0144	2.38 %

Step 3: do the three slopes agree?

Compare each pair using their combined uncertainty: $z = |m_i - m_j| / \sqrt{(\delta m_i)^2 + (\delta m_j)^2}$

1 vs 2: $0.0080 / 0.0267 = \mathbf{0.30}$; 1 vs 3: $0.0071 / 0.0241 = \mathbf{0.29}$; 2 vs 3: $0.0151 / 0.0234 = \mathbf{0.65}$. All are below 1, so the slopes agree within their fit uncertainties.

Step 4: g from each recording

Along the tilted table $a_y = -g \sin \theta$, so $g = |m| / \sin \theta$, with $\sin 3.5^\circ = 0.06105$. Two inputs are uncertain (m and θ), so their *relative* uncertainties add in quadrature:

$$\delta g / g = \sqrt{(\delta m / m)^2 + (\delta(\sin \theta) / \sin \theta)^2}, \quad \text{and since } \delta(\sin \theta) = \cos \theta \cdot \delta \theta, \quad \delta(\sin \theta) / \sin \theta = \delta \theta / \tan \theta$$

Angle term: $\delta \theta = 0.1^\circ \times \pi / 180 = 0.001745 \text{ rad}$ (must be in radians); $\tan 3.5^\circ = 0.06116$; $\delta \theta / \tan \theta = 0.001745 / 0.06116 = \mathbf{0.0285} = \mathbf{2.85 \%}$

Rec.	$g = m / \sin \theta (\text{m s}^{-2})$	$\delta m / m $	$\delta \theta / \tan \theta$	$\delta g / g = \sqrt{a^2 + b^2}$	$\delta g (\text{m s}^{-2})$	Result
1	$0.6133 / 0.06105 = 10.046$	3.16 %	2.85 %	$\sqrt{(3.16^2 + 2.85^2)} = 4.26 \%$	$10.046 \times 0.0426 = 0.43$	$\mathbf{10.05 \pm 0.43}$
2	$0.6213 / 0.06105 = 10.178$	2.96 %	2.85 %	$\sqrt{(2.96^2 + 2.85^2)} = 4.11 \%$	$10.178 \times 0.0411 = 0.42$	$\mathbf{10.18 \pm 0.42}$
3	$0.6062 / 0.06105 = 9.930$	2.38 %	2.85 %	$\sqrt{(2.38^2 + 2.85^2)} = 3.72 \%$	$9.930 \times 0.0372 = 0.37$	$\mathbf{9.93 \pm 0.37}$

Step 5: combine the three slopes, two ways

Mean slope: $\bar{a} = (-0.6133 + (-0.6213) + (-0.6062)) / 3 = -1.8409 / 3 = \mathbf{-0.6136 \text{ m s}^{-2}}$

A. Propagated from the fit errors. For a mean of three values, $\delta \bar{a} = \sqrt{(\delta m_1)^2 + (\delta m_2)^2 + (\delta m_3)^2} / 3 = \sqrt{(0.0194^2 + 0.0184^2 + 0.0144^2)} / 3 = \sqrt{0.000922} / 3 = 0.0304 / 3 = \mathbf{0.0101 \text{ m s}^{-2}}$

B. Statistical (standard error of the mean). Deviations from the mean: 0.00032, -0.00772, 0.00740; sample standard deviation $\sigma = \sqrt{[\sum (m_i - \bar{a})^2 / (3 - 1)]} = \sqrt{(0.000114 / 2)} = 0.00756$; SEM = $\sigma / \sqrt{3} = 0.00756 / 1.732 = \mathbf{0.0044 \text{ m s}^{-2}}$

Which one? A (0.0101) is about 2.3× larger than B (0.0044). Step 3 showed the slopes agree within their fit errors, so the scatter between recordings is no bigger than the random error already in each fit. B is small only because three values happened to land close together, and a standard deviation from just three numbers is itself very uncertain (roughly $\pm 50 \%$). So A is the more honest (more conservative) choice.

Step 6: combined g and its uncertainty

$$g = |\bar{a}| / \sin \theta = 0.6136 / 0.06105 = 10.051 \text{ m s}^{-2}$$

$$\delta g / g = \sqrt{(\delta \bar{a} / \bar{a})^2 + (\delta \theta / \tan \theta)^2} = \sqrt{(0.0101 / 0.6136)^2 + 0.0285^2} = \sqrt{(1.65^2 + 2.85^2)} \% = \mathbf{3.30 \%}$$

$\delta g = 10.051 \times 0.0330 = 0.331 \text{ m s}^{-2} \rightarrow g = (10.05 \pm 0.33) \text{ m s}^{-2} \approx (10.1 \pm 0.3) \text{ m s}^{-2}$ (uncertainty rounded to 1 s.f., value to the same decimal place)

With the SEM instead: $\sqrt{(0.71^2 + 2.85^2)} \% = 2.94 \% \rightarrow \pm 0.30 \text{ m s}^{-2}$. The angle term dominates either way.

Why not simply average g_1, g_2, g_3 and use $\sqrt{(\Sigma \delta g_i^2)/3}$? That gives $\pm 0.23 \text{ m s}^{-2}$, but it wrongly treats the angle error as independent in each recording. All three recordings used the same tilt, measured once, so its 2.85 % is a shared (systematic) error: it does not shrink when you average. That is why the slopes are averaged first and the angle uncertainty is added once at the end.

Step 7: compare with the tabulated value

$$z = |g - g_{\text{ref}}| / \delta g = |10.051 - 9.81| / 0.331 = 0.241 / 0.331 = 0.73$$

$z < 1$, so the result agrees with 9.81 m s^{-2} within one standard uncertainty ($|z| < 2$ is the usual threshold). Individual recordings: $z = 0.55, 0.88, 0.33$, all < 1 .

Step 8: are the error bars a sensible size?

Count the points whose error bar touches the fitted line, i.e. $|v - (mt + c)| \leq 0.085 \text{ m s}^{-1}$: recording 1: $35/78 = 45 \%$, recording 2: $46/65 = 71 \%$, recording 3: $50/76 = 66 \%$; overall $131/219 = 60 \%$. For 1σ error bars about 68 % should touch the line, so ± 2 mm is a realistic position uncertainty (slightly small for recording 1, whose video has a coarser scale).

Doing the same in Excel

Quantity	Excel formula (adjust the cell ranges)
Error bar δv_y	=SQRT(2)*0.002/(1/30)
Mid-point time	=(A7+A8)/2 (next to each v_y)
Slope m and δm	=INDEX(LINEST(vy_range, tmid_range, TRUE, TRUE), 1, 1) and =INDEX(LINEST(...), 2, 1)
g	=ABS(m)/SIN(RADIANS(3.5))
δg	=g*SQRT((dm/m)^2 + (RADIANS(0.1)/TAN(RADIANS(3.5)))^2)
Propagated $\delta \bar{a}$	=SQRT(SUMSQ(dm1, dm2, dm3))/3
SEM	=STDEV.S(m1, m2, m3)/SQRT(3)